# The Voyager Interstellar Mission

## Mission Overview

Both spacecraft were launched in 1977 to exploit a rare planetary alignment that occurs once every 176 years. Engineers called this opportunity the **grand tour trajectory**, because a single gravity-assist chain could reach all four outer planets in under twelve years.

Voyager 1 crossed the heliopause in August 2012, becoming the first human-made object to enter interstellar space. Its twin followed six years later at a different heliographic latitude.

## Scientific Instruments

Each spacecraft carries eleven investigations, most of which still return data on a 22.4-watt downlink.

### *The Low-Energy Charged Particle Detector*

The detector measures ions with energies between 22 keV and several hundred MeV using two solid-state telescope heads.